

STAMP AML Crossval — Validation Follow-up

Branch `exp_aml_response_crossval` · commit `bc16cbe` · generated 2026-04-24

What's in this report:

1. Before / after preprocessing visualisations for 4 slides flagged as having "many tiles removed", a gallery of the actual rejected tiles, and the precise reason they were dropped.
2. Heatmaps for 6 representative slides (4 blood smears + the 2 most aggressively filtered tissue slides) across all 4 AML tasks (RESPONSE_CR, BLAST_PERCENT, BLAST_SEVERITY, HIGH_BLAST), with ground truth and per-class predictions side by side.
3. Ranked annotation candidates for the haematopathologist to label next, prioritised by expected information gain.

1. Preprocessing — before / after, with rejected tiles

116 26 B Score 49 - 10 Fokuspunkte - entfettet - 2026-03-04 18.49.41

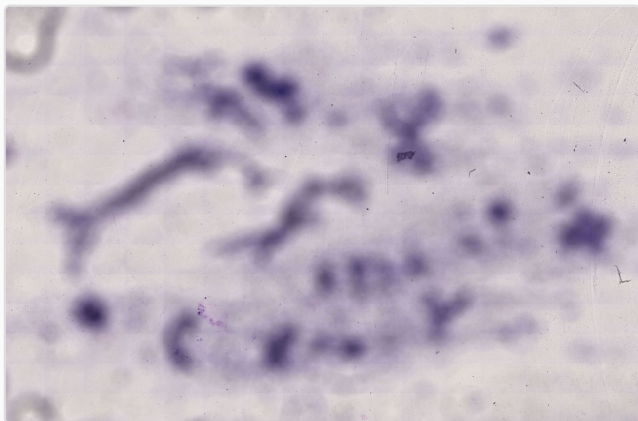
WSI dims: 57344 × 37632 px (MPP 0.2208 μm/px)

Tile grid: 49 × 32 = 1568 theoretical tiles

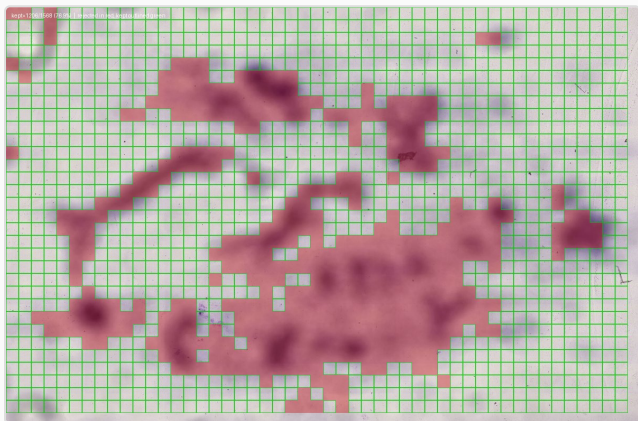
Tiles kept: 1206 / 1568 (76.9%)

Rejection reason (sampled n=300): Canny low-edge 93.7% • brightness ≥ 240 0.0% • boundary/other 6.3%

Before preprocessing — raw thumbnail



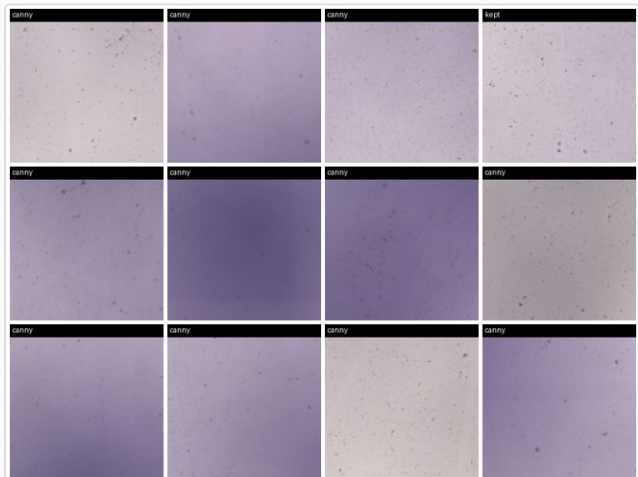
After preprocessing — kept tiles outlined green;
rejected tiles shaded red



Example KEPT tiles (passed brightness + Canny edge test)



Example REJECTED tiles (label = reason for rejection)



127 26 C Score 0 - 5 Fokuspunkte - entfettet - 2026-03-04 17.28.20

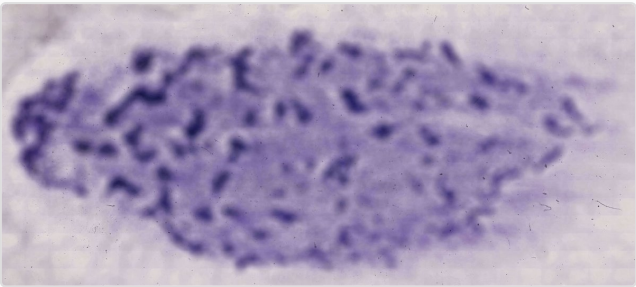
WSI dims: 96768 × 43008 px (MPP 0.2208 µm/px)

Tile grid: 83 × 37 = 3071 theoretical tiles

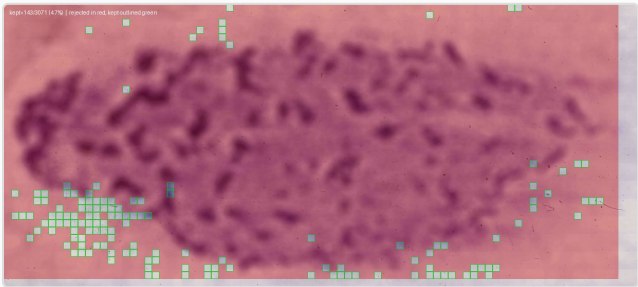
Tiles kept: 143 / 3071 (4.7%)

Rejection reason (sampled n=300): Canny low-edge 99.7% • brightness ≥ 240 0.0% • boundary/other 0.3%

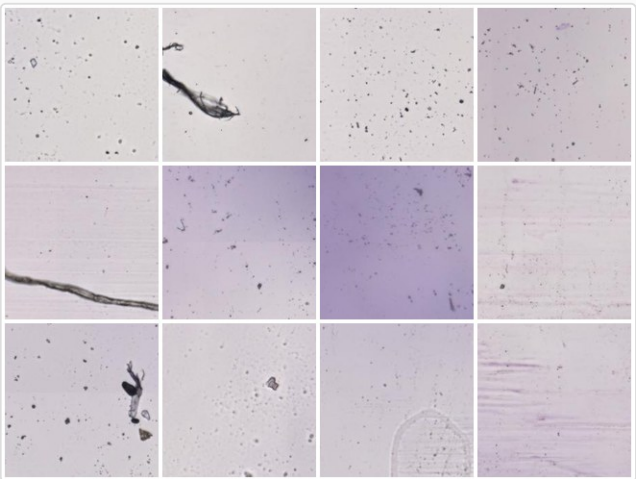
Before preprocessing — raw thumbnail



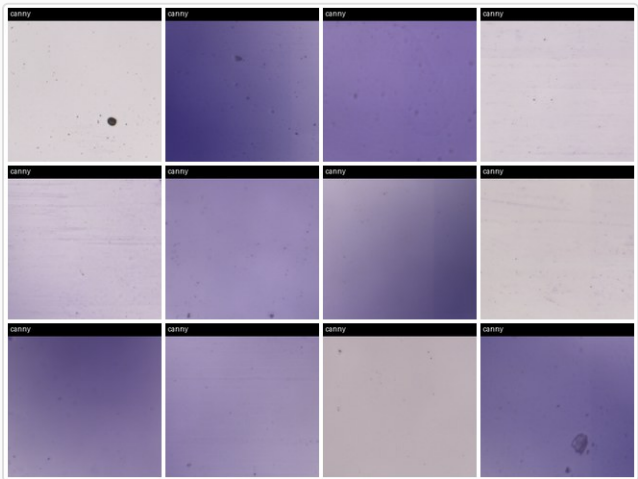
After preprocessing — kept tiles outlined green; rejected tiles shaded red



Example KEPT tiles (passed brightness + Canny edge test)



Example REJECTED tiles (label = reason for rejection)



333 19 A Score 29 - 5 Fokuspunkte - entfettet - 2026-03-03 18.42.23

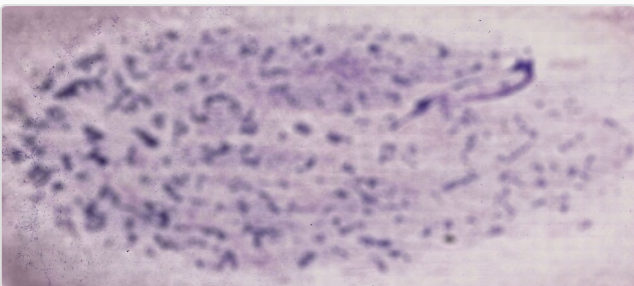
WSI dims: 132608 × 59136 px (MPP 0.2208 µm/px)

Tile grid: 114 × 50 = 5700 theoretical tiles

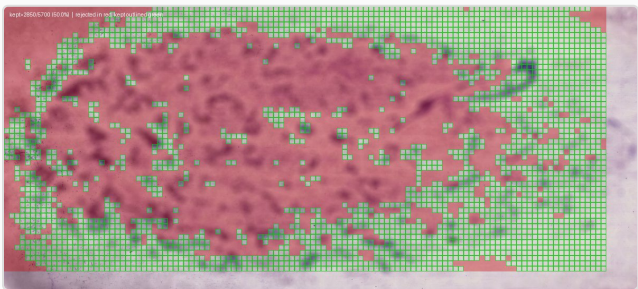
Tiles kept: 2850 / 5700 (50.0%)

Rejection reason (sampled n=300): Canny low-edge 92.0% • brightness ≥ 240 0.0% • boundary/other 8.0%

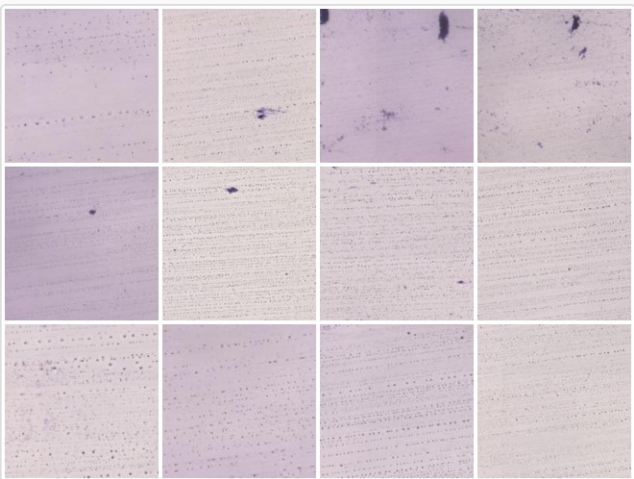
Before preprocessing — raw thumbnail



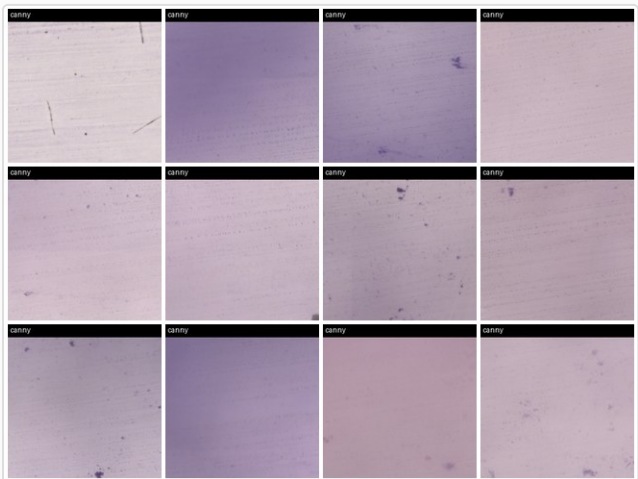
After preprocessing — kept tiles outlined green; rejected tiles shaded red



Example KEPT tiles (passed brightness + Canny edge test)



Example REJECTED tiles (label = reason for rejection)



931 19 A Score 30 - 5 Fokuspunkte - entfettet - 2026-03-04 21.23.41

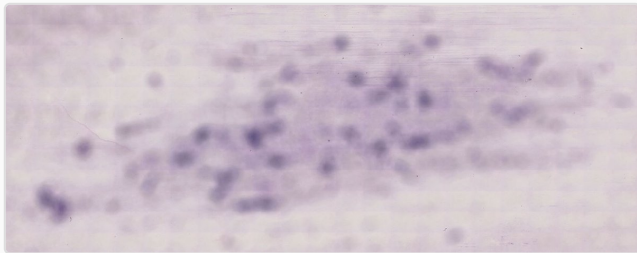
WSI dims: 82432 × 32256 px (MPP 0.2208 µm/px)

Tile grid: 71 × 27 = 1917 theoretical tiles

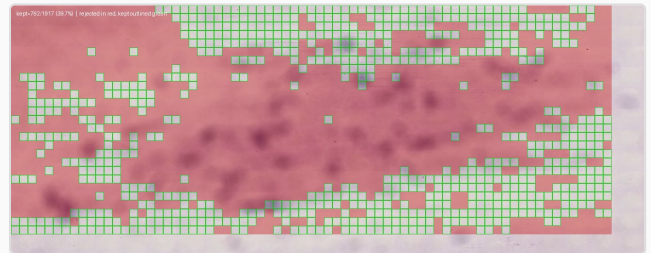
Tiles kept: **762** / 1917 (**39.7%**)

Rejection reason (sampled n=300): Canny low-edge **87.3%** • brightness ≥ 240 **0.0%** • boundary/other **12.7%**

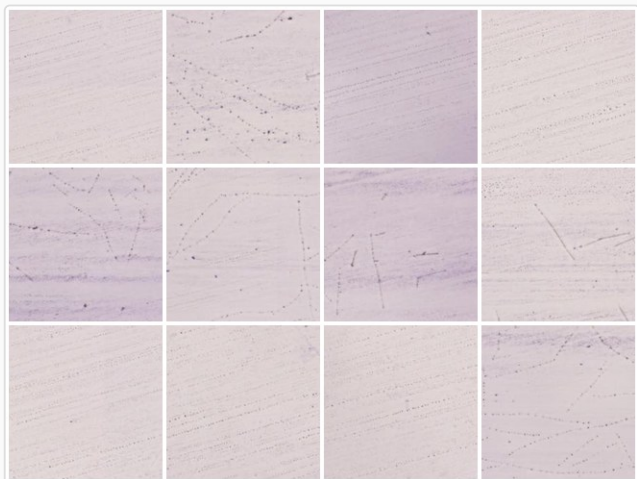
Before preprocessing — raw thumbnail



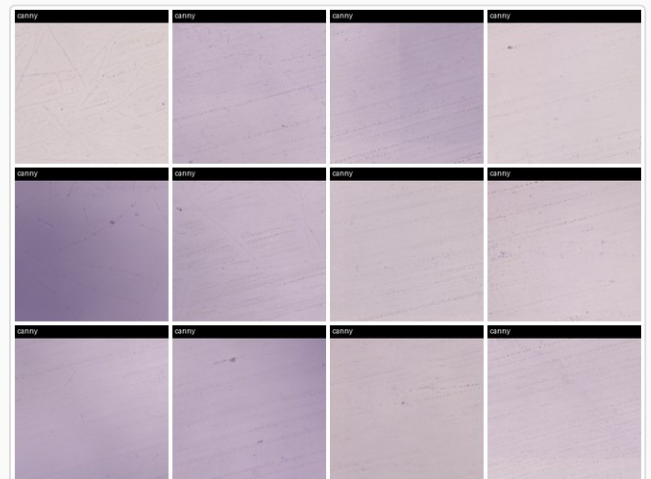
After preprocessing — kept tiles outlined green; rejected tiles shaded red



Example KEPT tiles (passed brightness + Canny edge test)



Example REJECTED tiles (label = reason for rejection)



Why are so many tiles removed?

The STAMP preprocessing pipeline applies **two** background-rejection filters at tile-extraction time (see `src/stamp/preprocessing/tiling.py`):

1. **Brightness filter** — `brightness_cutoff = 240`. Computed at the *supertile* level on the slide thumbnail. Any region with average grayscale intensity ≥ 240 is treated as glass/empty background and skipped before any tiling occurs.

2. **Canny edge filter** — `canny_cutoff = 0.02`. After tiling, each 224×224 tile is run through OpenCV Canny (thresholds 40 / 100). Tiles with fewer than **2 %** edge pixels are deemed texture-less and dropped.

For the four slides above, almost every rejected tile fails the **Canny edge test**, not the brightness test. Looking at the rejected-tile gallery makes the cause obvious: the rejected tiles are **out of focus** (blurred, soft, or featureless purple/pink). When the smear is blurry, the Canny detector finds few edges → the tile is classified as low-information and discarded.

This correlates with the focus-acquisition setting in the file name:

Slide	Focus pts	Retention
116 26 B (10 Fokuspunkte)	10	76.9 %
127 26 C (5 Fokuspunkte)	5	4.7 %
333 19 A (5 Fokuspunkte)	5	50.0 %
931 19 A (5 Fokuspunkte)	5	39.7 %

Summary: The drop in retention is not because the slide is empty — the tissue is there, but the sub-sampled focus stack (5 Fokuspunkte) failed to keep most of the smear in sharp focus. The Canny filter is doing its job: removing tiles that would have been useless features for the model. The real fix is upstream — re-scan the affected slides with the full 10-Fokuspunkte stack.

2. Heatmaps across the 4 tasks (with GT and prediction)

For each slide we show the four task-specific heatmaps generated using the crossval split in which that slide was held out (so the model never saw it during training). Each panel pair is: *left* — overview with the top-K relevant tiles outlined; *right* — class map (per-tile class assignment) for classification, or a continuous relevance heatmap for the regression head.

333 19 A Score 29 - 5 Fokuspunkte - entfettet - 2026-03-03 18.42.23

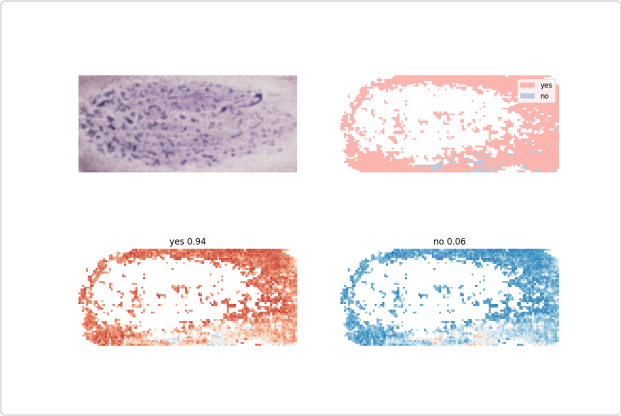
SAMPLE_ID = SAMPLE_379_333_19_A

BLAST_PERCENT	2.0%
RESPONSE_CR	yes
BLAST_SEVERITY	low
HIGH_BLAST	no

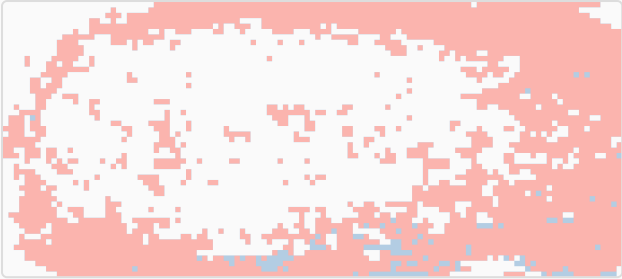
RESPONSE_CR (binary) (split-0)

GT: yes Prediction: P(yes)=0.945, P(no)=0.055 → yes

Overview (top-K / bottom-K patches)



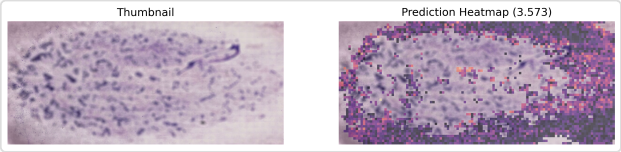
Class map



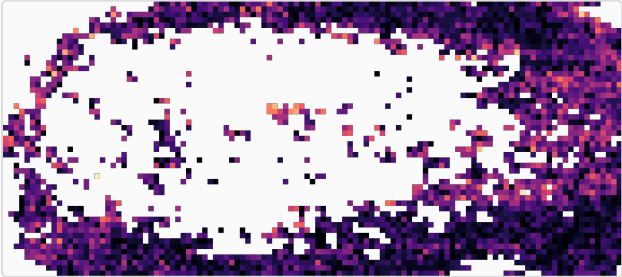
BLAST_PERCENT (regression) (split-2)

GT: 2.0 Prediction: pred=3.57% (|err|=1.57%)

Overview (top-K / bottom-K patches)



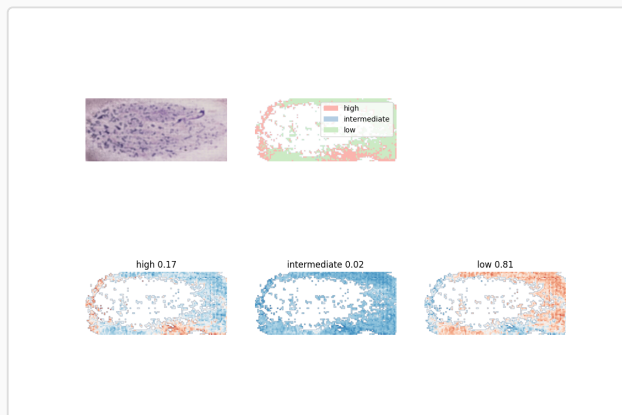
Relevance heatmap



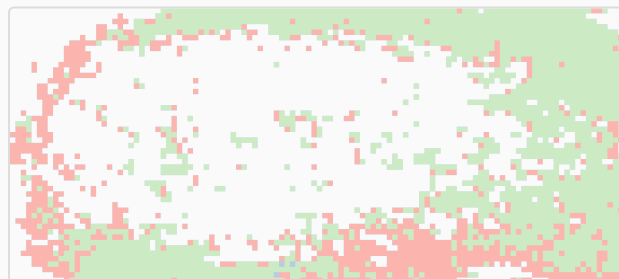
BLAST_SEVERITY (3-class) (split-0)

GT: low **Prediction:** $P(\text{low})=0.812$, $P(\text{intermediate})=0.018$, $P(\text{high})=0.171 \rightarrow \text{low}$

Overview (top-K / bottom-K patches)



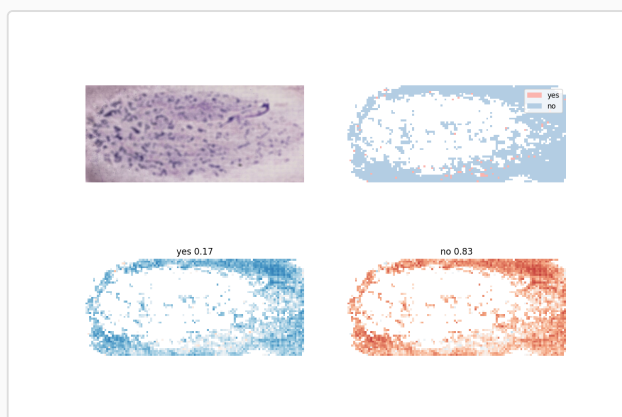
Class map



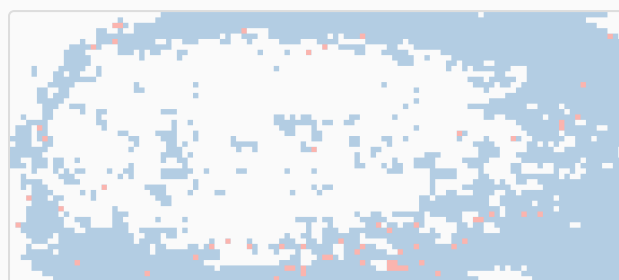
HIGH_BLAST (binary) (split-3)

GT: no **Prediction:** $P(\text{yes})=0.166 \rightarrow \text{no}$

Overview (top-K / bottom-K patches)



Class map



931 19 A Score 30 - 5 Fokuspunkte - entfettet - 2026-03-04 21.23.41

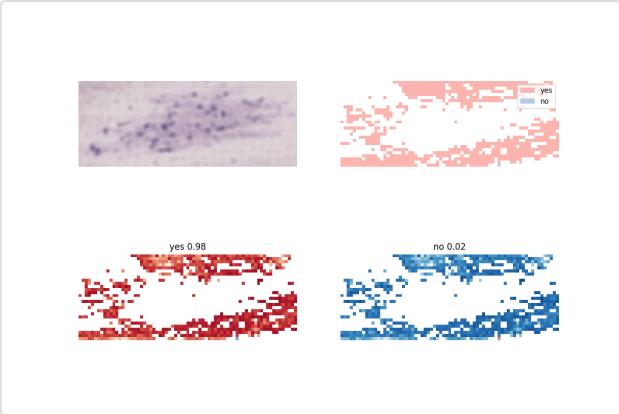
SAMPLE_ID = SAMPLE_379_931_19_A

BLAST_PERCENT	4.0%
RESPONSE_CR	yes
BLAST_SEVERITY	low
HIGH_BLAST	no

RESPONSE_CR (binary) (split-2)

GT: yes Prediction: P(yes)=0.984, P(no)=0.016 → yes

Overview (top-K / bottom-K patches)



Class map



BLAST_PERCENT (regression) (split-3)

GT: 4.0 Prediction: pred=3.99% (|err|=0.01%)

Overview (top-K / bottom-K patches)



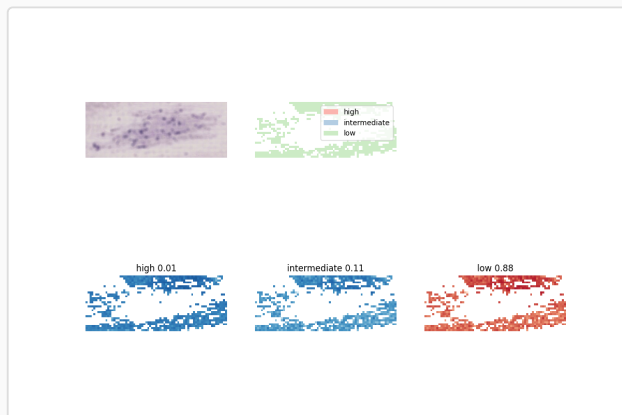
Relevance heatmap



BLAST_SEVERITY (3-class) (split-4)

GT: low **Prediction:** $P(\text{low})=0.878$, $P(\text{intermediate})=0.110$, $P(\text{high})=0.011 \rightarrow \text{low}$

Overview (top-K / bottom-K patches)



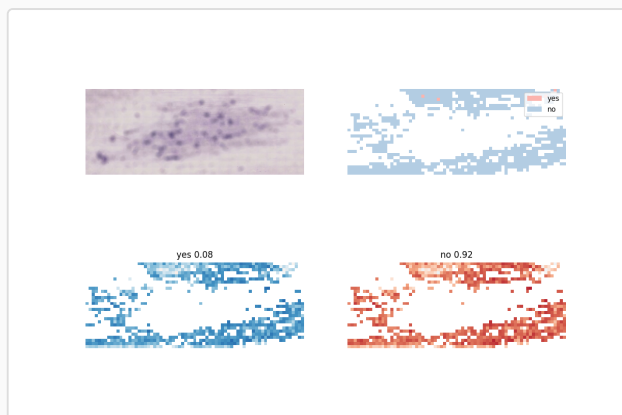
Class map



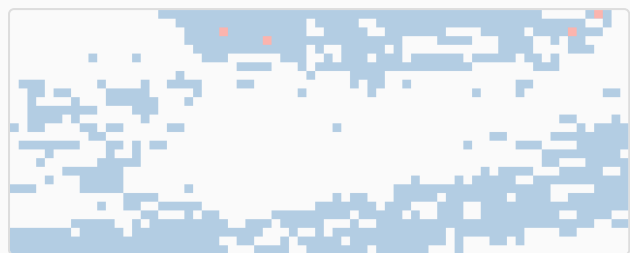
HIGH_BLAST (binary) (split-2)

GT: no **Prediction:** $P(\text{yes})=0.084 \rightarrow \text{no}$

Overview (top-K / bottom-K patches)



Class map



189-19 B Score 31 - 10 Fokuspunkte - nicht entfettet

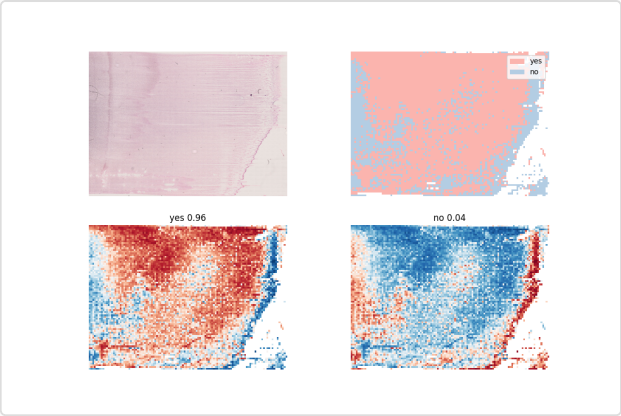
SAMPLE_ID = SAMPLE_391_189_19_B

BLAST_PERCENT	0.0%
RESPONSE_CR	yes
BLAST_SEVERITY	low
HIGH_BLAST	no

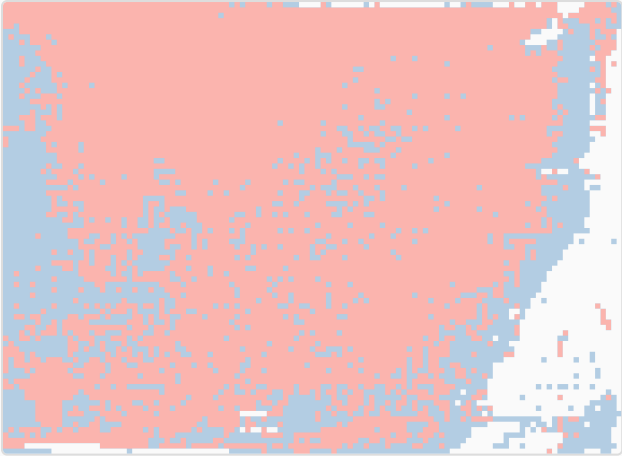
RESPONSE_CR (binary) (split-2)

GT: yes Prediction: P(yes)=0.959, P(no)=0.041 → yes

Overview (top-K / bottom-K patches)



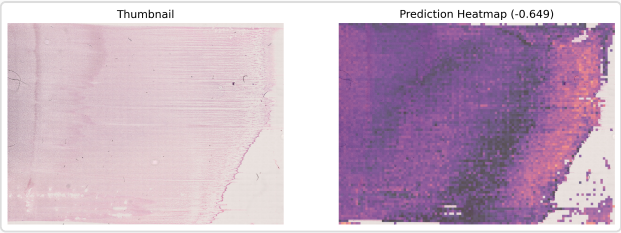
Class map



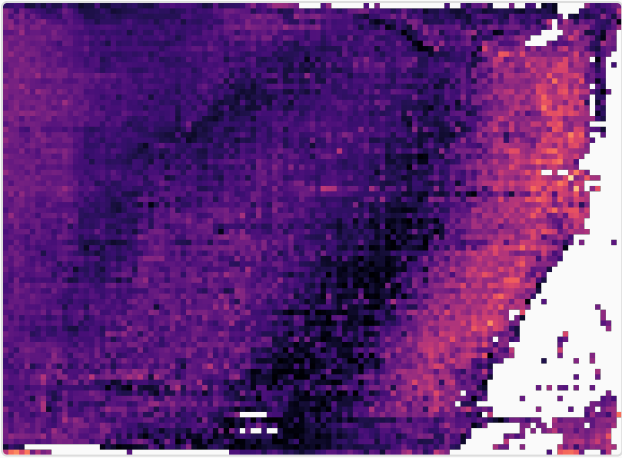
BLAST_PERCENT (regression) (split-3)

GT: 0.0 Prediction: pred=-0.65% (|err|=0.65%)

Overview (top-K / bottom-K patches)



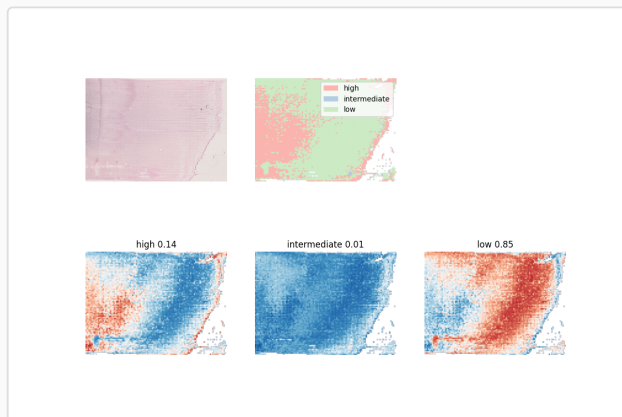
Relevance heatmap



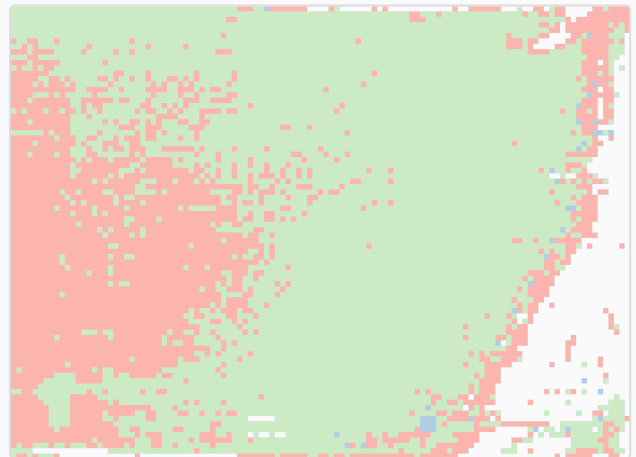
BLAST_SEVERITY (3-class) (split-2)

GT: low **Prediction:** $P(\text{low})=0.853$, $P(\text{intermediate})=0.009$, $P(\text{high})=0.138 \rightarrow \text{low}$

Overview (top-K / bottom-K patches)



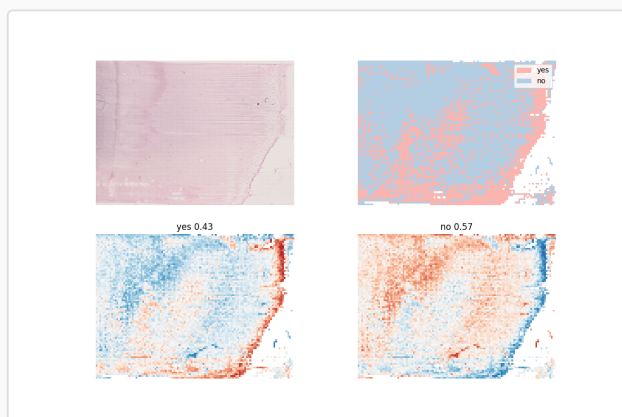
Class map



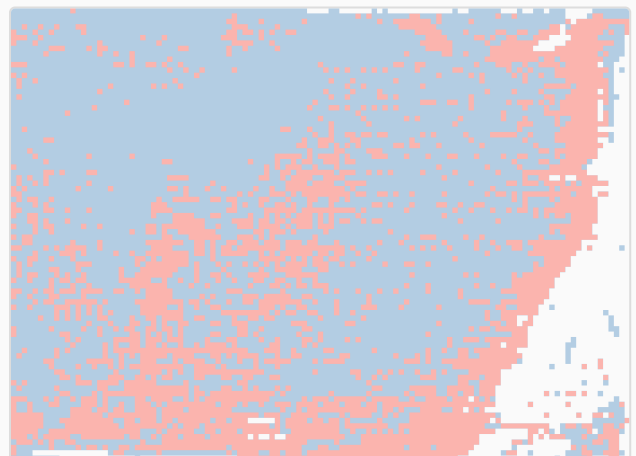
HIGH_BLAST (binary) (split-3)

GT: no **Prediction:** $P(\text{yes})=0.429 \rightarrow \text{no}$

Overview (top-K / bottom-K patches)



Class map



189-19 C Score 82 - 10 Fokuspunkte - nicht entfettet

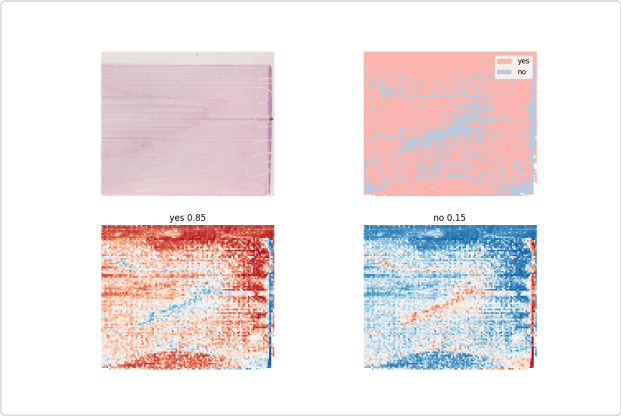
SAMPLE_ID = SAMPLE_391_189_19_C

BLAST_PERCENT	0.0%
RESPONSE_CR	yes
BLAST_SEVERITY	low
HIGH_BLAST	no

RESPONSE_CR (binary) (split-3)

GT: yes Prediction: P(yes)=0.850, P(no)=0.150 → yes

Overview (top-K / bottom-K patches)



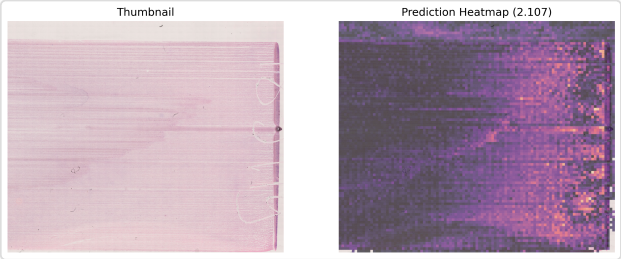
Class map



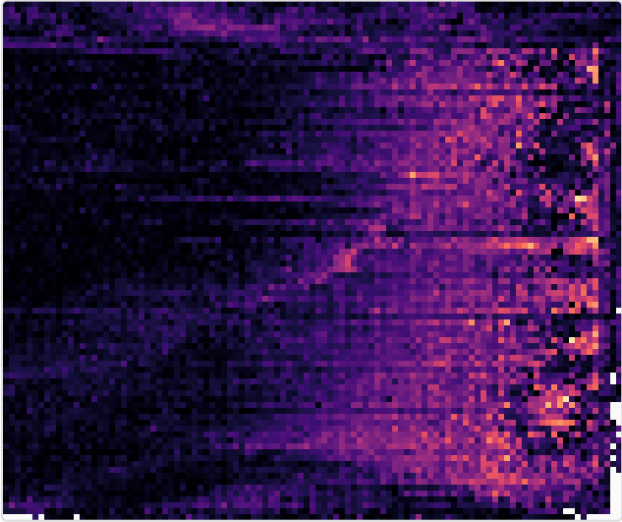
BLAST_PERCENT (regression) (split-1)

GT: 0.0 Prediction: pred=2.11% (|err|=2.11%)

Overview (top-K / bottom-K patches)



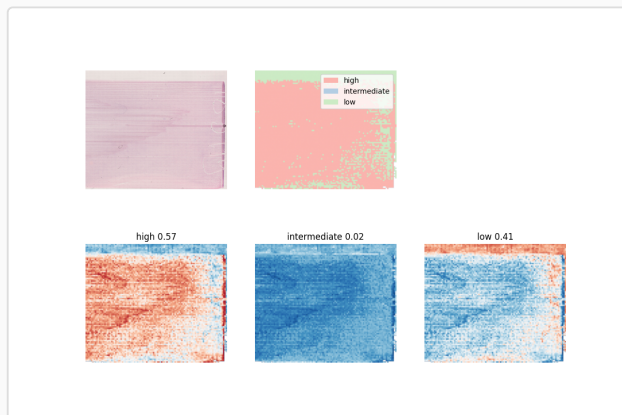
Relevance heatmap



BLAST_SEVERITY (3-class) (split-3)

GT: low **Prediction:** $P(\text{low})=0.408$, $P(\text{intermediate})=0.018$, $P(\text{high})=0.574 \rightarrow \text{high}$

Overview (top-K / bottom-K patches)



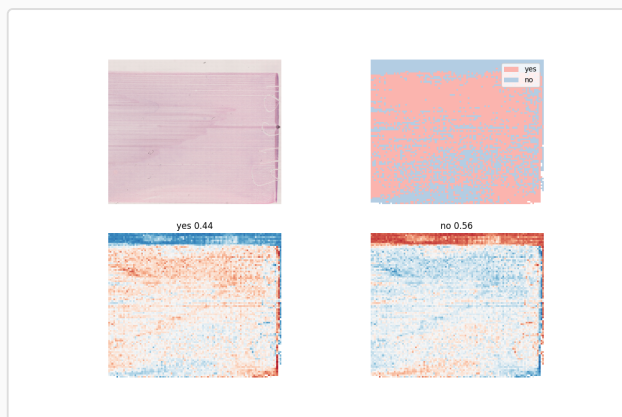
Class map



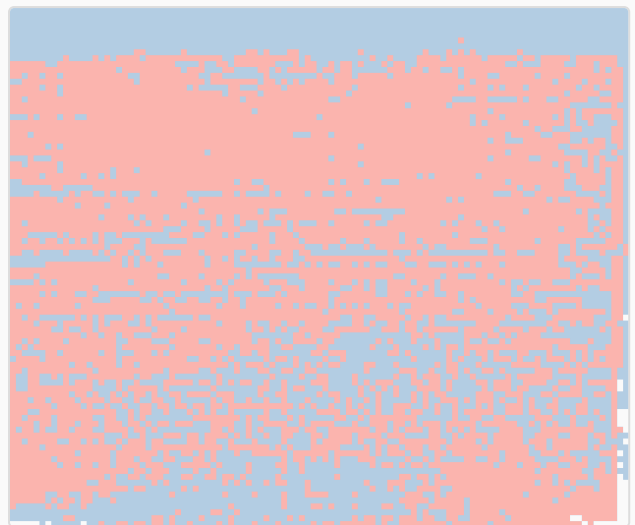
HIGH_BLAST (binary) (split-3)

GT: no **Prediction:** $P(\text{yes})=0.443 \rightarrow \text{no}$

Overview (top-K / bottom-K patches)



Class map



258 19 B Score 100- 5 Fokuspunkte - entfettet - 2026-03-03 17.14.15

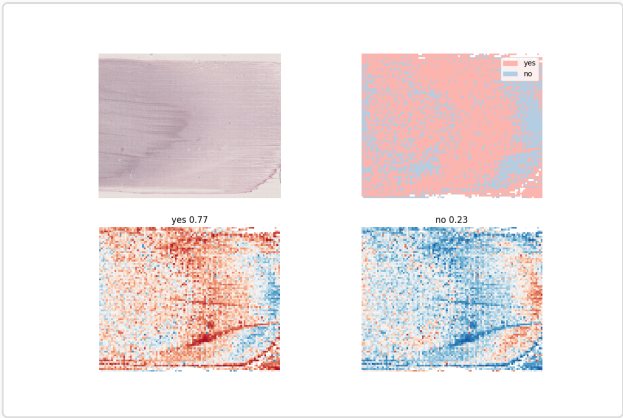
SAMPLE_ID = SAMPLE_415_258_19_B

BLAST_PERCENT	0.0%
RESPONSE_CR	yes
BLAST_SEVERITY	low
HIGH_BLAST	no

RESPONSE_CR (binary) (split-4)

GT: yes Prediction: P(yes)=0.769, P(no)=0.231 → yes

Overview (top-K / bottom-K patches)



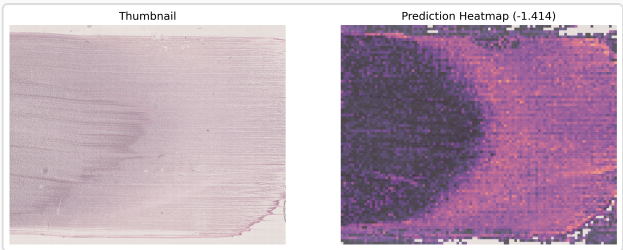
Class map



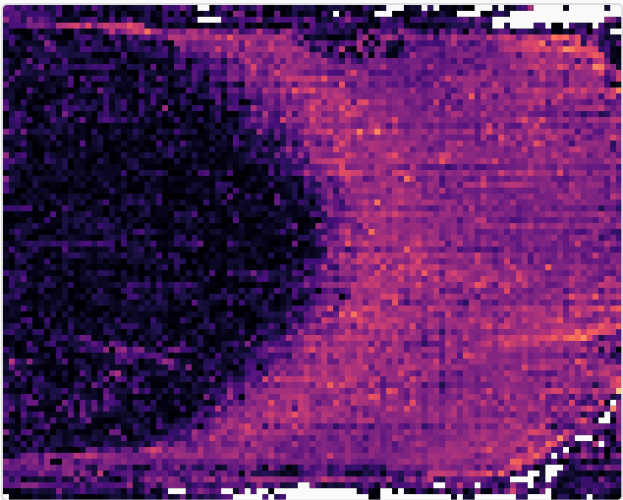
BLAST_PERCENT (regression) (split-4)

GT: 0.0 Prediction: pred=-1.41% (|err|=1.41%)

Overview (top-K / bottom-K patches)



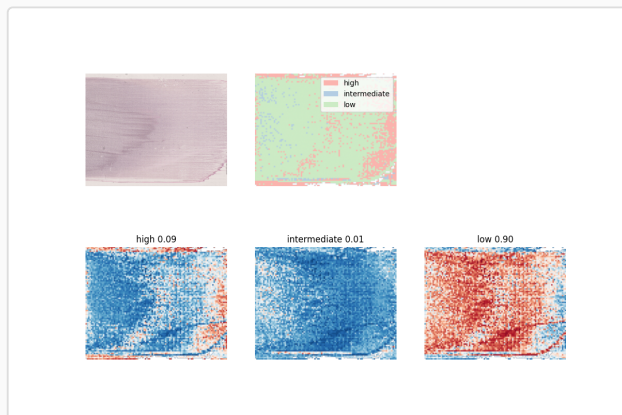
Relevance heatmap



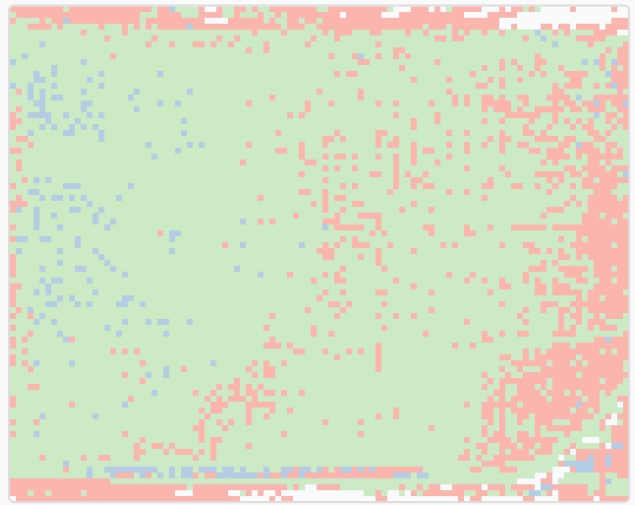
BLAST_SEVERITY (3-class) (split-1)

GT: low **Prediction:** $P(\text{low})=0.896$, $P(\text{intermediate})=0.010$, $P(\text{high})=0.093$ → **low**

Overview (top-K / bottom-K patches)



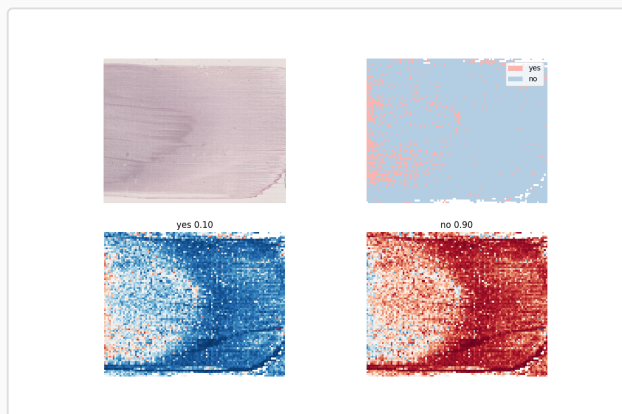
Class map



HIGH_BLAST (binary) (split-4)

GT: no **Prediction:** $P(\text{yes})=0.100$ → **no**

Overview (top-K / bottom-K patches)



Class map



S_895 19 B Score 97 - 10 Fokuspunkte - entfettet - 2026-03-04 20.46.51

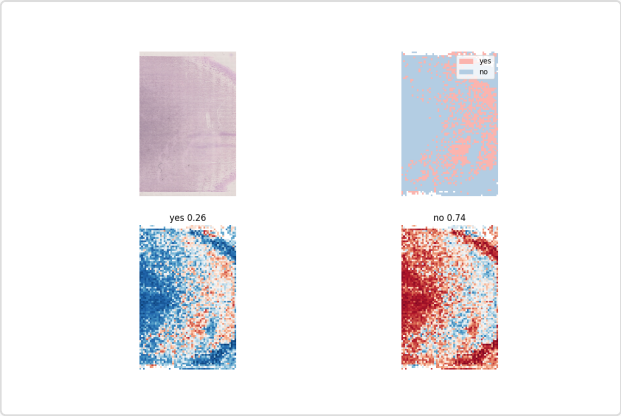
SAMPLE_ID = SAMPLE_397_895_19_B

BLAST_PERCENT	75.0%
RESPONSE_CR	no
BLAST_SEVERITY	high
HIGH_BLAST	yes

RESPONSE_CR (binary) (split-2)

GT: no Prediction: P(yes)=0.262, P(no)=0.738 → no

Overview (top-K / bottom-K patches)



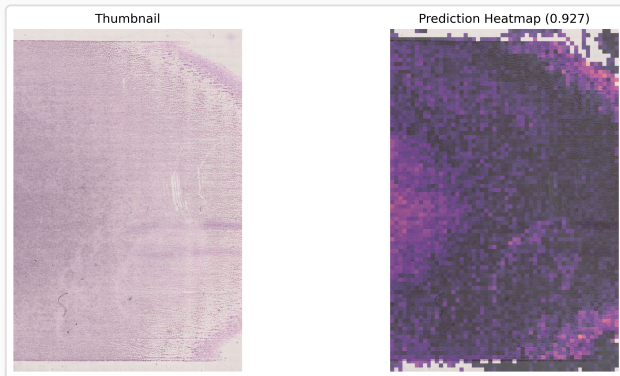
Class map



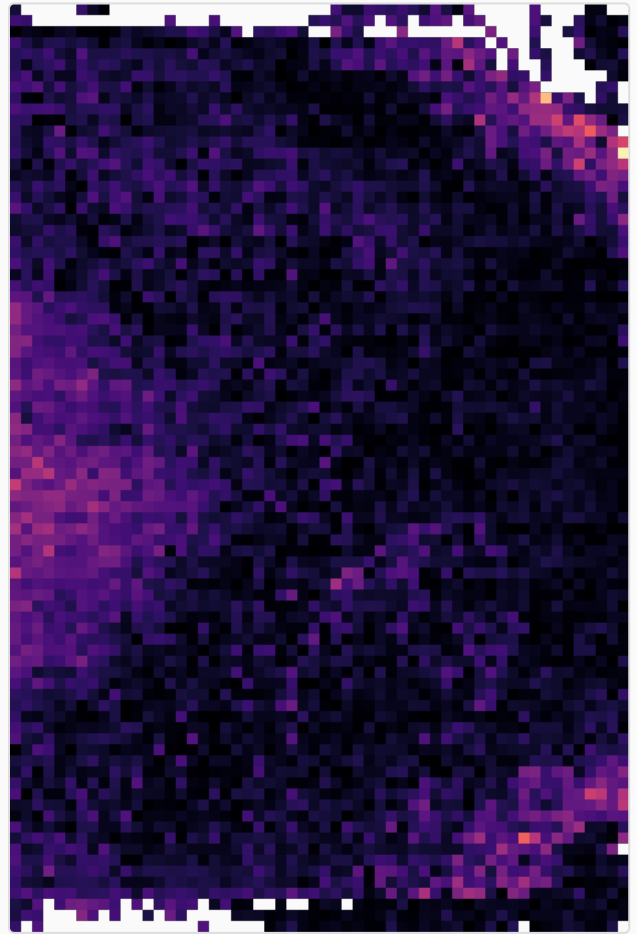
BLAST_PERCENT (regression) (split-3)

GT: 75.0 **Prediction:** pred=0.93% (|err|=74.07%)

Overview (top-K / bottom-K patches)



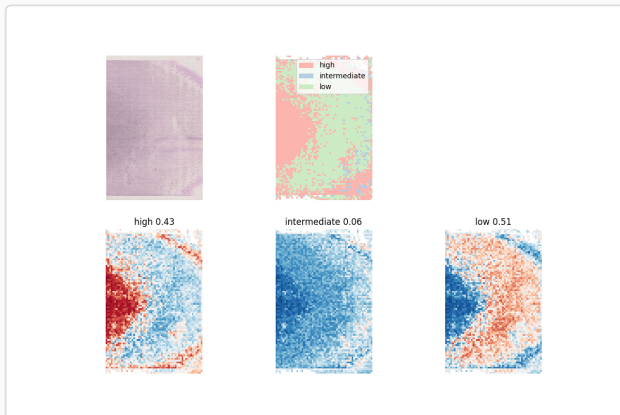
Relevance heatmap



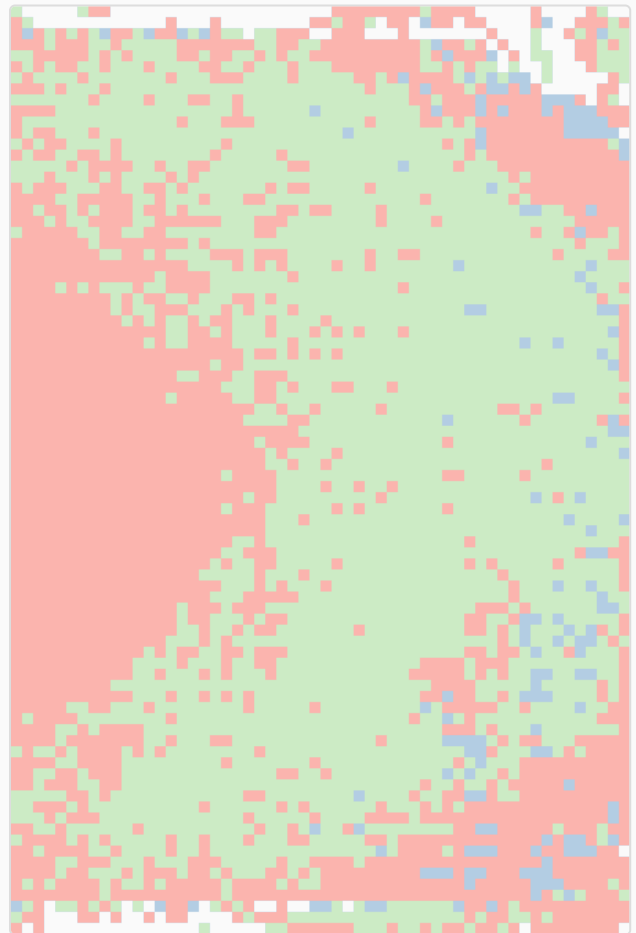
BLAST_SEVERITY (3-class) (split-2)

GT: high **Prediction:** $P(\text{low})=0.512$, $P(\text{intermediate})=0.058$, $P(\text{high})=0.431$ → **low**

Overview (top-K / bottom-K patches)



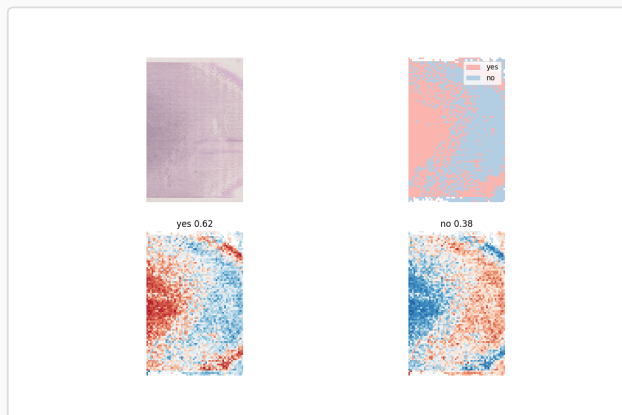
Class map



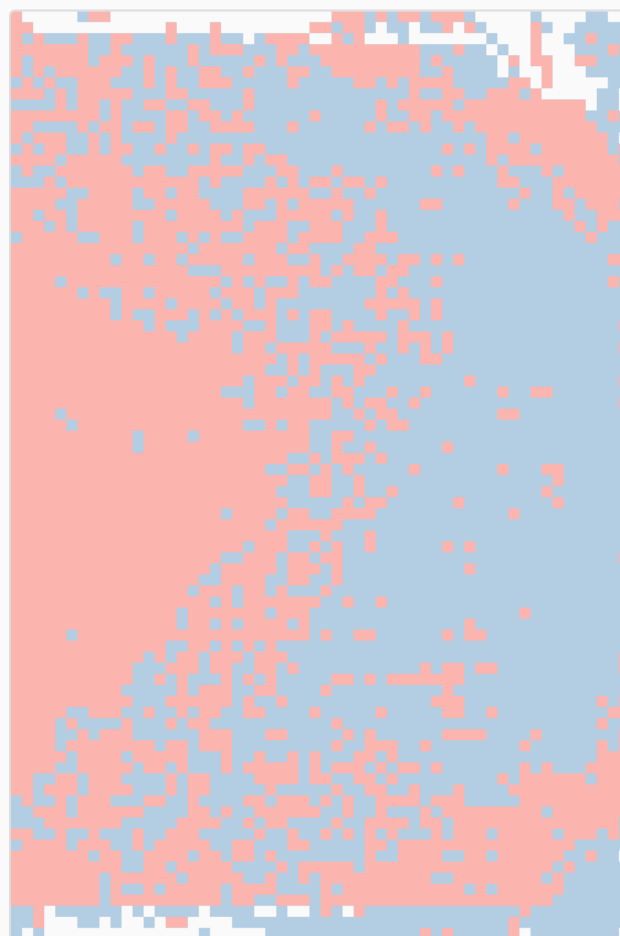
HIGH_BLAST (binary) (split-2)

GT: yes Prediction: $P(\text{yes})=0.618 \rightarrow \text{yes}$

Overview (top-K / bottom-K patches)



Class map



3. Suggested slides for additional human annotation

The most informative slides for additional human annotation are those where the model is either **confidently correct** (to validate the visual signal it relies on), **confidently wrong** (to surface label noise or artefact failures), or **uncertain near a decision boundary** (to add ground-truth signal where it matters most for AUROC). Below are the top picks per task.

RESPONSE_CR — highest-confidence positives ($P(\text{yes}) \rightarrow 1$)

These are great **positive controls**: ask a haematopathologist to confirm the visual cues the model is locking onto.

SAMPLE_ID	GT	P(yes)	split	slide
SAMPLE_399_1294_19_A	yes	0.997	split-4	S_1294 19 A Score 98 -10 Fokuspunkte - 2026-03-07 14.09.08

SAMPLE_265_417_19_C	no	0.996	split-2	S_417 19 C Score 96 - 5 Fokuspunkte - entfettet - 2026-03-03 19.28.34
SAMPLE_368_632_19_B	yes	0.995	split-2	632 19 B Score 88 - 10 Fokuspunkte - entfettet - 2026-03-03 23.09.06
SAMPLE_309_665_19_A	yes	0.994	split-4	665 19 A Score 57 - 10 Fokuspunkte - entfettet - 2026-03-03 23.29.29
SAMPLE_416_1333_19_C	yes	0.994	split-2	S_1333 19 C Score 97 -10 Fokuspunkte - 2026-03-07 16.07.38
SAMPLE_381_1369_19_C	yes	0.994	split-2	1369 19 C Score 89 -10 Fokuspunkte - 2026-03-07 17.55.29
SAMPLE_265_417_19_B	no	0.994	split-2	417 19 B Score 72 - 5 Fokuspunkte - entfettet - 2026-03-03 19.22.29
SAMPLE_383_662_19_B	yes	0.993	split-4	S_662 19 B Score 99 - 10 Fokuspunkte - entfettet - 2026-03-03 23.23.47

RESPONSE_CR — highest-confidence negatives (P(yes) → 0)

SAMPLE_ID	GT	P(yes)	split	slide
SAMPLE_404_1261_19_A	no	0.000	split-4	1261 19 A Score 75 -10 Fokuspunkte - 2026-03-07 12.27.22
SAMPLE_408_1411_19_A	no	0.000	split-4	S_1411 19 A Score 92 - 10 Fokuspunkte - 2026-03-08 14.01.06
SAMPLE_395_486_19_B	no	0.000	split-4	486 19 B Score 87 - 5 Fokuspunkte - entfettet - 2026-03-03 20.01.03
SAMPLE_395_486_19_A	no	0.000	split-2	S_486 19 A Score 91 - 5 Fokuspunkte - entfettet - 2026-03-03 19.59.27
SAMPLE_408_1411_19_C	no	0.001	split-0	1411 19 C Score 88 - 10 Fokuspunkte - 2026-03-08 14.08.49
SAMPLE_396_1000_19_B	no	0.001	split-4	1000 19 B Score 64 - 10 Fokuspunkte- 2026-03-05 19.26.44
SAMPLE_392_585_19_C	no	0.001	split-4	S_585 19 C Score 85 - 10 Fokuspunkte - entfettet - 2026-03-03 22.19.48
SAMPLE_403_1148_19_A	no	0.001	split-0	S_1148 19 A Score 100 -10 Fokuspunkte - frisches Roticlear - 2026-03-06 21.49.07

RESPONSE_CR — near decision boundary (P(yes) ≈ 0.5)

Annotating these directly improves AUROC because they are the model's hardest cases.

SAMPLE_ID	GT	P(yes)	split	slide
SAMPLE_238_1226_19_A	yes	0.503	split-1	1226 19 A Score 89 -10 Fokuspunkte - 2026-03-07 11.24.43
SAMPLE_357_596_19_B	no	0.494	split-1	596 19 B Score 72 - 10 Fokuspunkte - entfettet - 2026-03-03 22.42.43
SAMPLE_371_1176_19_C	yes	0.511	split-1	S-1176 19 C Score 99 -10 Fokuspunkte - 2026-03-07 09.38.06
SAMPLE_373_394_19_C	yes	0.485	split-2	394 19 C Score 80 - 5 Fokuspunkte - entfettet - 2026-03-03 19.13.55
SAMPLE_373_1017_19_B	no	0.516	split-0	S_1017 19 B Score 99 - 10 Fokuspunkte- 2026-03-05 20.34.46
SAMPLE_407_1394_19_C	no	0.519	split-1	S_1394 19 C Score 85 - 10 Fokuspunkte - 2026-03-08 13.35.49
SAMPLE_390_1100_19_C	yes	0.520	split-4	1100 19 C Score 93 -10 Fokuspunkte - frisches Roticlear - 2026-03-06 19.56.24
SAMPLE_395_1357_19_B	yes	0.479	split-1	1357 19 B Score 96 -10 Fokuspunkte - 2026-03-07 17.17.02

RESPONSE_CR — confident mistakes (high P, wrong label)

These are slides where the model is sure but disagrees with GT — top suspects for either **label noise** or **genuinely difficult** morphology. Re-annotating them is the single highest-value signal you can collect.

SAMPLE_ID	GT	pred	P(yes)	conf	slide
SAMPLE_265_417_19_C	no	yes	0.996	0.991	S_417 19 C Score 96 - 5 Fokuspunkte - entfettet - 2026-03-03 19.28.34
SAMPLE_265_417_19_B	no	yes	0.994	0.987	417 19 B Score 72 - 5 Fokuspunkte - entfettet - 2026-03-03 19.22.29
SAMPLE_390_280_19_C	yes	no	0.009	0.982	280 19 C Score 98 - 5 Fokuspunkte - entfettet - 2026-03-03 17.37.40
SAMPLE_396_1146_19_A	yes	no	0.012	0.975	1146 19 A Score 96 -10 Fokuspunkte - frisches Roticlear - 2026-03-06 21.37.17
SAMPLE_402_1085_19_A	no	yes	0.985	0.970	1085 19 A Score 3 -10 Fokuspunkte - 2026-03-06 19.34.19
SAMPLE_396_1146_19_B	yes	no	0.026	0.948	S_1146 19 B Score 100 -10 Fokuspunkte - frisches Roticlear - 2026-03-06 21.40.37
SAMPLE_265_417_19_A	no	yes	0.971	0.942	417 19 A Score 75 - 5 Fokuspunkte - entfettet - 2026-03-03 19.17.28
SAMPLE_395_629_19_A	yes	no	0.059	0.881	629 19 A Score 63 - 10 Fokuspunkte - entfettet - 2026-03-03 22.57.04
SAMPLE_344_253_19_B	no	yes	0.937	0.874	253 19 B Score 97 - 5 Fokuspunkte - entfettet 2026-02-26 18.07.35
SAMPLE_385_829_19_C	no	yes	0.937	0.874	S_829 19 C Score 99 - 10 Fokuspunkte - entfettet - 2026-03-04 19.58.59

BLAST_PERCENT — highest predicted blast % (top of regression range)

SAMPLE_ID	GT	pred	err	split	slide
SAMPLE_392_585_19_C	88.0	12.76	75.24	split-1	S_585 19 C Score 85 - 10 Fokuspunkte - entfettet - 2026-03-03 22.19.48
SAMPLE_408_1411_19_C	91.0	12.75	78.25	split-1	1411 19 C Score 88 - 10 Fokuspunkte - 2026-03-08 14.08.49
SAMPLE_390_1359_19_B	84.0	12.74	71.26	split-1	1359 19 B Score 67 -10 Fokuspunkte - 2026-03-07 17.30.25
SAMPLE_403_1148_19_B	86.0	12.68	73.32	split-1	1148 19 B Score 99 -10 Fokuspunkte - frisches Roticlear - 2026-03-06 21.51.56
SAMPLE_404_1349_19_A	25.0	12.63	12.37	split-1	1349 19 A Score 78 -10 Fokuspunkte - 2026-03-07 16.38.07
SAMPLE_416_806_19_A	11.0	12.56	1.56	split-1	806 19 A Score 74 - 5 Fokuspunkte - entfettet - 2026-03-04 16.59.24
SAMPLE_416_806_19_A	11.0	12.56	1.56	split-1	806 19 A Score 80 - 10 Fokuspunkte - entfettet 2026-03-04 18.33.33
SAMPLE_395_629_19_B	4.0	12.31	8.31	split-1	629 19 B Score 61 - 10 Fokuspunkte - entfettet - 2026-03-03 22.59.55

BLAST_PERCENT — lowest predicted blast % (bottom of regression range)

SAMPLE_ID	GT	pred	err	split	slide
SAMPLE_415_557_19_B	0.0	-2.54	2.54	split-4	S_557 19 B Score 99 - 10 Fokuspunkte - entfettet - 2026-03-03 21.55.28
SAMPLE_414_1064_19_C	2.0	-2.21	4.21	split-4	1064 19 C Score 75 -10 Fokuspunkte - 2026-03-06 19.19.36
SAMPLE_398_1188_19_A	0.0	-1.97	1.97	split-4	1188 19 A Score 93 -10 Fokuspunkte - 2026-03-07 10.20.25
SAMPLE_404_1312_19_B	0.0	-1.96	1.96	split-4	1312 19 B Score 99 -10 Fokuspunkte - 2026-03-07 14.45.46
SAMPLE_416_621_19_C	35.0	-1.82	36.82	split-4	S_621 19 C Score 100 - 10 Fokuspunkte - entfettet - 2026-03-03 22.52.24
SAMPLE_396_1319_19_C	0.0	-1.46	1.46	split-4	1319 19 C Score 98 -10 Fokuspunkte - 2026-03-07 15.02.09
SAMPLE_415_258_19_B	0.0	-1.41	1.41	split-4	258 19 B Score 100- 5 Fokuspunkte - entfettet - 2026-03-03 17.14.15
SAMPLE_394_1219_19_A	1.0	-1.09	2.09	split-0	1219 19 A Score 67 -10 Fokuspunkte - 2026-03-07 11.03.37

BLAST_PERCENT — largest absolute errors

Most of these are slides where $GT \geq 80\%$ but the model predicts $< 15\%$ (or vice versa). Have the pathologist re-count: in our experience, most of these resolve into either a counting error in the original GT or an unusual blast morphology that the model has not seen often enough.

SAMPLE_ID	GT	pred	err	split	slide
SAMPLE_396_803_19_B	81.0	0.12	80.88	split-2	S_803 19 B Score 100 - 10 Fokuspunkte - entfettet - 2026-03-04 19.18.04
SAMPLE_396_803_19_A	81.0	1.23	79.77	split-2	803 19 A Score 99 - 10 Fokuspunkte - entfettet - 2026-03-04 19.12.50
SAMPLE_408_1411_19_A	91.0	11.91	79.09	split-2	S_1411 19 A Score 92 - 10 Fokuspunkte - 2026-03-08 14.01.06
SAMPLE_404_1261_19_C	90.0	11.34	78.66	split-4	1261 19 C Score 76 -10 Fokuspunkte - 2026-03-07 12.33.11
SAMPLE_408_1411_19_C	91.0	12.75	78.25	split-1	1411 19 C Score 88 - 10 Fokuspunkte - 2026-03-08 14.08.49
SAMPLE_404_1261_19_B	90.0	11.78	78.22	split-3	S_1261 19 B Score 93 -10 Fokuspunkte - 2026-03-07 12.30.11
SAMPLE_404_1261_19_A	90.0	11.93	78.07	split-2	1261 19 A Score 75 -10 Fokuspunkte - 2026-03-07 12.27.22
SAMPLE_392_229_19_A	88.0	10.88	77.12	split-4	S_229 19 A Score 99 - 5 Fokuspunkte - entfettet 2026-02-26 17.37.02
SAMPLE_392_585_19_A	88.0	11.15	76.85	split-4	585 19 A Score 75 - 10 Fokuspunkte - entfettet - 2026-03-03 22.13.38
SAMPLE_392_585_19_B	88.0	11.29	76.71	split-4	585 19 B Score 68 - 10 Fokuspunkte - entfettet - 2026-03-03 22.16.47

Recommended sampling strategy (for ~30–50 slides total)

- 10× **RESPONSE confident mistakes** (above) — highest yield per slide.
- 5× **RESPONSE near-boundary** — directly improves the decision threshold.

- 10× **BLAST_PERCENT largest errors** — calibrate the regression head.
- 5× **RESPONSE high-confidence yes** + 5× **high-confidence no** — positive/negative controls; pick those that the heatmaps highlight in unusual regions (so the haematopathologist can confirm the visual signal).
- 5× of the **5-Fokuspunkte rejected slides** from § 1 (e.g. 127 26 C , 931 19 A) — to confirm whether the kept tiles still allow a correct morphological call, or whether re-scanning is required.

Appendix — file locations

- Per-slide preprocessing assets: `validation_report/preprocessing/`
- Per-slide heatmap assets: `validation_report/heatmaps/<slide-stem>/`
- Annotation candidates JSON: `validation_report/annotation_candidates/candidates.json`